ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2022 P2H Worked Solutions

Jan 2022 | P2H

33 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 4

TOPIC: **Expanding Brackets**

Jan 2022 P2H - Jan 2022 | P2H

1 (a) Expand and simplify $(y + 4)(2 - y)$

....
(2)

(b) Factorise fully $15b^5c - 35b^3c^9$

....
(2)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Expanding Brackets**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. (a)

(a)

$$(y+4)(2-y) = 2y - y^2 + 8 - 4y = -y^2 - 2y + 8$$

2. (b)

(b)

$$15b^5c - 35b^3c^9 = 5b^3c(3b^2 - 7c^8)$$

FINAL ANSWER

(a) $-y^2 - 2y + 8$, (b) $5b^3c(3b^2 - 7c^8)$

PAST PAPER QUESTION Q#: 2 Marks: 3

TOPIC: **Fractions**

Jan 2022 P2H - Jan 2022 | P2H

2 Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Fractions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Convert to improper fractions

Convert to improper fractions:

$$6\frac{3}{4} = \frac{27}{4}$$

$$2\frac{4}{7} = \frac{18}{7}$$

2. Divide by multiplying by the

Divide by multiplying by the reciprocal:

$$\frac{27}{4} \div \frac{18}{7} = \frac{27}{4} \times \frac{7}{18}$$

$$= \frac{21}{8} = 2\frac{5}{8}$$

FINAL ANSWER

$2\frac{5}{8}$

PAST PAPER QUESTION

Q#: 3 Marks: 3

TOPIC: **Congruence, Similarity & Geometrical Proof**

Jan 2022 P2H - Jan 2022 | P2H

3

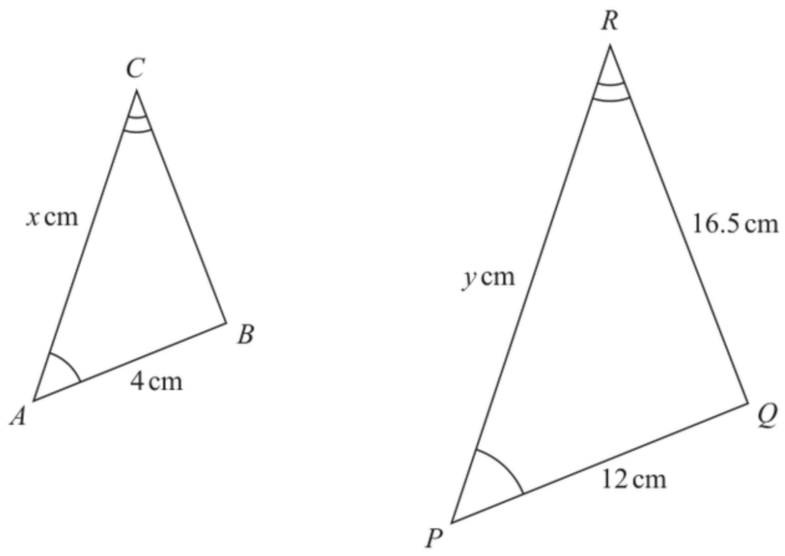

Diagram **NOT** accurately drawn

Triangle ABC is similar to triangle PQR

$AB = 4$ cm $\quad PQ = 12$ cm $\quad RQ = 16.5$ cm $\quad AC = x$ cm $\quad PR = y$ cm

(a) Calculate the length of BC

.... cm

(2)

(b) Write down an expression for y in terms of x

$y =$ ....

(1)

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Congruence, Similarity & Geometrical Proof**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. The scale factor from triangle

The scale factor from triangle ABC to triangle PQR is

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

2. Evaluate fraction

So

$$BC = \frac{RQ}{3} = \frac{16.5}{3} = 5.5$$

3. Calculate value

Also PR corresponds to AC, so

$$y = 3x$$

FINAL ANSWER

$BC = 5.5$ cm, and $y = 3x$.

PAST PAPER QUESTION

Q#: 4 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P2H - Jan 2022 | P2H

4 Each side of a regular octagon has a length of 18 mm, correct to the nearest 0.5 mm

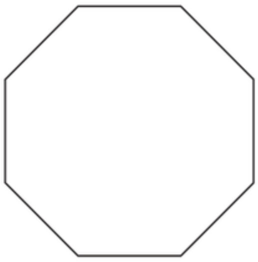

Diagram **NOT** accurately drawn

(a) Write down the lower bound of the length of each side of the octagon.

.... mm

(1)

(b) Write down the upper bound of the length of each side of the octagon.

.... mm

(1)

(Total for Question 4 is 2 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. The side length is mm

The side length is 18 mm correct to the nearest 0.5 mm.

2. Half of mm is mm

Half of 0.5 mm is 0.25 mm.

$$18 - 0.25 = 17.75$$

$$18 + 0.25 = 18.25$$

FINAL ANSWER

(a) 17.75 mm, (b) 18.25 mm

PAST PAPER QUESTION Q#: 5 Marks: 4

TOPIC: **Bearings, Scale Drawing & Constructions**

Jan 2022 P2H - Jan 2022 | P2H

5 The scale diagram shows the position on a map of a house, *A*

Scale: 1 cm represents 200 metres

House *C* is on a bearing of 110° from *A*
The distance from *A* to *C* is 700 m

(a) Mark the position of *C* on the diagram with a cross (×)
Label your cross *C*

(3)

(b) Write the scale of the map in the form 1 : *n*

1 : ....

(1)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION Q#: 5 Marks: 4

TOPIC: **Bearings, Scale Drawing & Constructions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. The distance from to is

The distance from A to C is 700 m.

2. The scale is

The scale is

$$1 \text{ cm represents } 200 \text{ m}$$

3. Use bearings

So the map distance is

$$700 \div 200 = 3.5 \text{ cm}$$

4. Use bearings

Part (a): from A, draw a bearing of 110° and mark C at 3.5 cm from A.

5. Use bearings

For part (b),

$$200 \text{ m} = 20000 \text{ cm}$$

6. Use bearings

So the scale is

$$1 : 20000$$

FINAL ANSWER

Draw C 3.5 cm from A on a bearing of 110°, and the scale is $1 : 20000$.

PAST PAPER QUESTION

Q#: 6 Marks: 5

TOPIC: **Probability Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

6 A bag contains only pink sweets, white sweets, green sweets and red sweets.

The table gives each of the probabilities that, when a sweet is taken at random from the bag, the sweet will be green or the sweet will be red.

Sweet	pink	white	green	red
Probability			0.2	0.35

The ratio
number of pink sweets : number of white sweets = 2 : 1

There are 28 red sweets in the bag.

Work out the number of white sweets in the bag.

(Total for Question 6 is 5 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 5

TOPIC: **Probability Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Split the ratio

The probabilities for green and red are

$$0.20 + 0.35 = 0.55$$

2. Split the ratio

So

$$P(\text{pink or white}) = 1 - 0.55 = 0.45$$

3. Split the ratio

Pink : white $= 2 : 1$, so white is $\frac{1}{3}$ of 0.45.

$$P(\text{white}) = \frac{1}{3} \times 0.45 = 0.15$$

4. Split the ratio

The red probability is 0.35, and there are 28 red sweets.

$$\text{total sweets} = \frac{28}{0.35} = 80$$

5. Number of white sweets

Number of white sweets:

$$0.15 \times 80 = 12$$

FINAL ANSWER

12

PAST PAPER QUESTION

Q#: 7 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

Jan 2022 P2H - Jan 2022 | P2H

7 Find the lowest common multiple (LCM) of 28, 42 and 63
Show your working clearly.

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Find the LCM

$$28 = 2^2 \times 7$$

$$42 = 2 \times 3 \times 7$$

$$63 = 3^2 \times 7$$

2. Find the LCM

For the LCM, take the highest power of each prime:

$$\text{LCM} = 2^2 \times 3^2 \times 7 = 252$$

FINAL ANSWER

252

PAST PAPER QUESTION Q#: 8 Marks: 5

TOPIC: **Percentages**

Jan 2022 P2H - Jan 2022 | P2H

8 The table gives information about the average house price in England in 2018 and in 2019

Year	**2017**	**2018**	**2019**
Average house price (£)		228 314	231 776

(a) Work out the percentage increase in the average house price from 2018 to 2019
Give your answer correct to one decimal place.

.... %
(2)

The average house price in 2019 was 7.7% greater than the average house price in 2017

(b) Work out the average house price in 2017
Give your answer correct to 3 significant figures.

£ ....
(3)

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION Q#: 8 Marks: 5

TOPIC: **Percentages**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Increase from 2018 to 2019

Increase from 2018 to 2019:

$$231776 - 228314 = 3462$$

2. Percentage increase

Percentage increase:

$$\frac{3462}{228314} \times 100 = 1.516\ldots$$

3. Correct to one decimal place

Correct to one decimal place:

$$1.5\%$$

4. Calculate percentage

For part (b), the 2019 price was 7.7% greater than the 2017 price.

$$231776 = 1.077 \times 2017 \text{ price}$$

$$2017 \text{ price} = \frac{231776}{1.077} = 215205.19\ldots$$

5. Correct to 3 significant figures

Correct to 3 significant figures:

$$215000$$

FINAL ANSWER

(a) 1.5%, (b) £215000

PAST PAPER QUESTION

Q#: 9 Marks: 4

TOPIC: **Statistics Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

9 The frequency table gives information about the number of points scored by a player.

Number of points	**Frequency**
0	13
1	17
2	8
3	x
4	11

The mean number of points scored is 2

Work out the value of x

$x =$ ....

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 4

TOPIC: **Statistics Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. The total number of points

The total number of points is

$$0(13) + 1(17) + 2(8) + 3x + 4(11) = 77 + 3x$$

2. The total frequency is

The total frequency is

$$13 + 17 + 8 + x + 11 = 49 + x$$

3. Calculate statistic

The mean is 2, so

$$\frac{77 + 3x}{49 + x} = 2$$

$$77 + 3x = 98 + 2x$$

$$x = 21$$

FINAL ANSWER

$x = 21$.

PAST PAPER QUESTION

Q#: 10 Marks: 3

TOPIC: **Simultaneous Equations**

Jan 2022 P2H - Jan 2022 | P2H

10 Solve the simultaneous equations

$$3x + 5y = 3.1$$
$$6x + 3y = 3.75$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Simultaneous Equations**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Solve simultaneous equations

$$3x + 5y = 3.1$$

$$6x + 3y = 3.75$$

2. Double the first equation

Double the first equation:

$$6x + 10y = 6.2$$

3. Subtract the second equation

Subtract the second equation:

$$7y = 2.45$$

$$y = 0.35$$

4. Solve simultaneous equations

Substitute into $3x + 5y = 3.1$:

$$3x + 5(0.35) = 3.1$$

$$3x + 1.75 = 3.1$$

$$3x = 1.35$$

$$x = 0.45$$

FINAL ANSWER

$x = 0.45$, $y = 0.35$.

PAST PAPER QUESTION

Q#: 11 **Marks: 4**

TOPIC: **Angles in Polygons & Parallel Lines**

Jan 2022 P2H - Jan 2022 | P2H

11 The diagram shows a regular 10-sided polygon, *ABCDEFGHIJ*

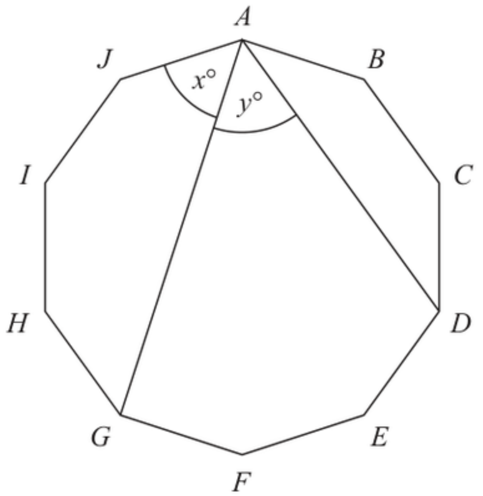

Diagram **NOT** accurately drawn

Show that $x = y$

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. A regular 10 sided polygon

A regular 10-sided polygon can be divided into 10 equal arcs around its circumcircle.

2. The angle subtends the arc

The angle x subtends the arc from J to G, which covers 3 equal sides of the decagon.

3. The angle subtends the arc

The angle y subtends the arc from G to D, which also covers 3 equal sides of the decagon.

4. Equal arcs subtend equal angles

Equal arcs subtend equal angles at the circumference, so

$$x = y$$

FINAL ANSWER

$x = y$.

PAST PAPER QUESTION

Q#: 12 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Jan 2022 P2H - Jan 2022 | P2H

12 $a = 6 \times 10^{40}$

Work out the value of a^3
Give your answer in standard form.

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Convert standard form

$$a = 6 \times 10^{40}$$

$$a^3 = (6 \times 10^{40})^3$$

$$a^3 = 6^3 \times 10^{120}$$

$$a^3 = 216 \times 10^{120}$$

2. Write this in standard form

Write this in standard form:

$$216 \times 10^{120} = 2.16 \times 10^{122}$$

FINAL ANSWER

2.16×10^{122}

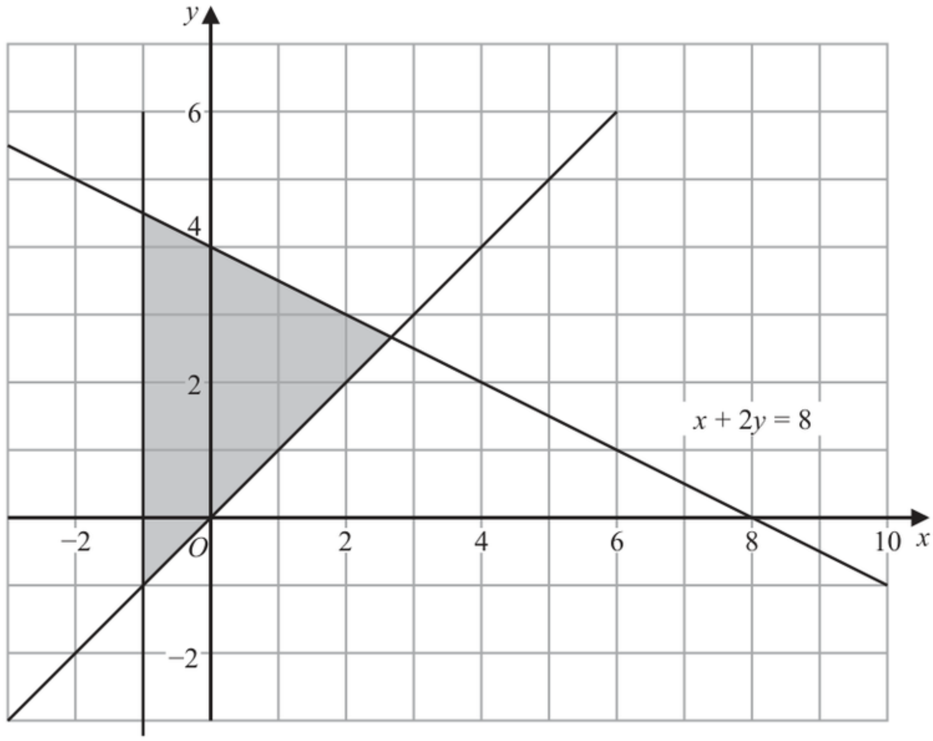
PAST PAPER QUESTION

Q#: 13 Marks: 3

TOPIC: **Graphing Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

13 The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down three inequalities that define the shaded region.

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Graphing Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. The region is to the

The region is to the right of the y-axis:

$$x \geq 0$$

2. It is above the line

It is above the line $y = x$:

$$y \geq x$$

3. It is below the line

It is below the line

$$x + 2y = 8$$

so

$$x + 2y \leq 8$$

FINAL ANSWER

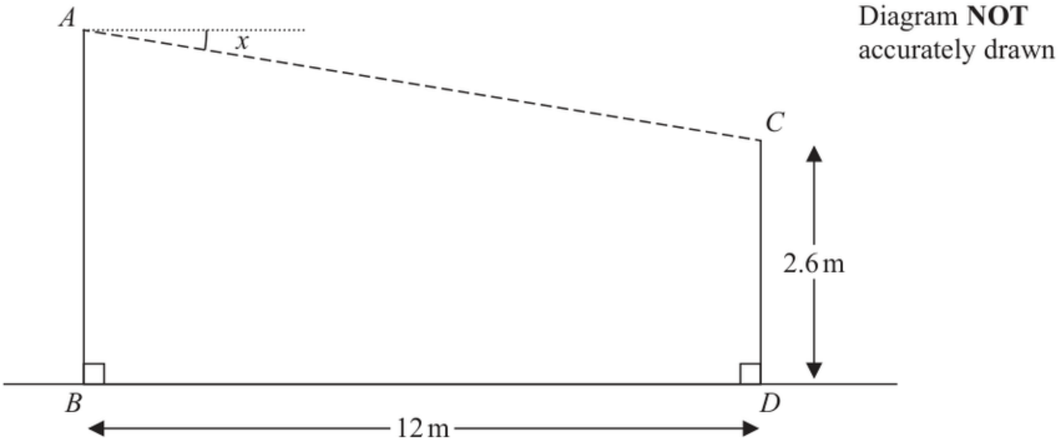
$x \geq 0$, $y \geq x$, $x + 2y \leq 8$.

PAST PAPER QUESTION Q#: 14 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P2H - Jan 2022 | P2H

14 A zip wire is shown as the dashed line AC in the diagram.



The zip wire is supported by two vertical posts AB and CD standing on horizontal ground.

$CD = 2.6\,\text{m}$ $\quad BD = 12\,\text{m}$

The zip wire makes an angle x with the horizontal, as shown in the diagram.
The design of the zip wire requires the angle x to be at least 5°

Work out the least possible height of the post AB
Give your answer correct to 3 significant figures.

.... m

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Use trigonometry

The horizontal distance is 12 m and the zip wire is 13 m.

2. Vertical difference

Vertical difference:

$$\sqrt{13^2 - 12^2} = 5$$

3. Use trigonometry

Since the shorter post is 2.6 m,

$$AB = 2.6 + 5 = 7.6$$

FINAL ANSWER

7.60 m.

PAST PAPER QUESTION

Q#: 15 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

15 Diyar recorded the distance, in kilometres, that he cycled each day for 11 days.
Here are his results.

8	10	12	13	5	23	21	7	5	16	14

Find the interquartile range of his results.

.... km

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Put the distances in order

Put the distances in order:

$$5,\ 5,\ 7,\ 8,\ 10,\ 12,\ 13,\ 14,\ 16,\ 21,\ 23$$

2. Calculate statistic

There are 11 values, so the median is the 6th value.

3. The lower half is

The lower half is

$$5,\ 5,\ 7,\ 8,\ 10$$

4. Calculate value

so

$$Q_1 = 7$$

5. The upper half is

The upper half is

$$13,\ 14,\ 16,\ 21,\ 23$$

6. Calculate value

so

$$Q_3 = 16$$

$$\text{IQR} = 16 - 7 = 9$$

FINAL ANSWER

9 km.

PAST PAPER QUESTION

Q#: 16 Marks: 4

TOPIC: **Circle Theorems**

Jan 2022 P2H - Jan 2022 | P2H

16 *D*, *E*, *F* and *G* are points on a circle, centre *O*

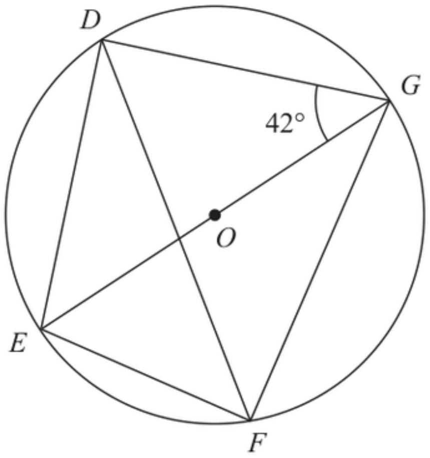

Diagram **NOT** accurately drawn

EOG is a diameter of the circle.

Angle $EGD = 42°$

Calculate the size of angle *DFG*
Give a reason for each stage of your working.

Angle *DFG* = ... °

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Circle Theorems**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Use trigonometry

Since EOG is a diameter,

$$\angle EDG = 90^\circ$$

2. Use trigonometry

In triangle EDG,

$$\angle DEG = 180^\circ - 90^\circ - 42^\circ = 48^\circ$$

3. Angles and stand on the

Angles DEG and DFG stand on the same chord DG, so they are equal.

$$\angle DFG = 48^\circ$$

FINAL ANSWER

48°.

PAST PAPER QUESTION

Q#: 17 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P2H - Jan 2022 | P2H

17 Show that $\dfrac{\sqrt{12}}{\sqrt{3}+2}$

can be written in the form $a + \sqrt{b}$ where a and b are integers.

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Simplify surd

$$\frac{\sqrt{12}}{\sqrt{3}+2} = \frac{2\sqrt{3}}{\sqrt{3}+2}$$

2. Multiply by the conjugate

Multiply by the conjugate:

$$\frac{2\sqrt{3}}{\sqrt{3}+2} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2\sqrt{3}(2-\sqrt{3})}{4-3}$$

$$= 4\sqrt{3} - 6$$

3. Simplify surd

Since $4\sqrt{3} = \sqrt{48}$,

$$4\sqrt{3} - 6 = -6 + \sqrt{48}$$

FINAL ANSWER

$-6 + \sqrt{48}$, so $a = -6$ and $b = 48$.

PAST PAPER QUESTION

Topic: **Algebraic Proof**

Jan 2022 P2H - Jan 2022 | P2H

Q#: 18 Marks: 3

18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Let the two consecutive odd

Let the two consecutive odd numbers be

$$2n+1 \quad \text{and} \quad 2n+3$$

2. The sum of their squares

The sum of their squares is

$$(2n+1)^2+(2n+3)^2$$

$$=4n^2+4n+1+4n^2+12n+9$$

$$=8n^2+16n+10$$

$$=8(n^2+2n+1)+2=8(n+1)^2+2$$

3. Use Algebraic proof correct

This is a multiple of 8 plus 2, so the remainder is always 2.

FINAL ANSWER

The remainder is always 2.

PAST PAPER QUESTION

Q#: 19 Marks: 3

TOPIC: **Circle Theorems**

Jan 2022 P2H - Jan 2022 | P2H

19

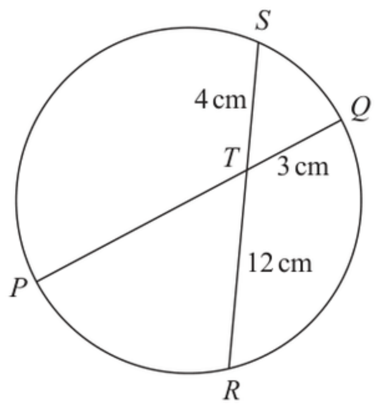

Diagram **NOT** accurately drawn

PTQ is a diameter of a circle.
RTS is a chord of the circle.

$TQ = 3$ cm $\qquad ST = 4$ cm $\qquad TR = 12$ cm

Calculate the radius of the circle.

.... cm

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION Q#: 19 Marks: 3

TOPIC: **Circle Theorems**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Use the intersecting chords theorem

Use the intersecting chords theorem:

$$PT \times TQ = ST \times TR$$

$$PT \times 3 = 4 \times 12$$

$$PT = 16$$

2. Use circle theorem

Since PTQ is a diameter,

$$PQ = PT + TQ = 16 + 3 = 19$$

3. Use circle theorem

So the radius is

$$19 \div 2 = 9.5$$

FINAL ANSWER

9.5 cm.

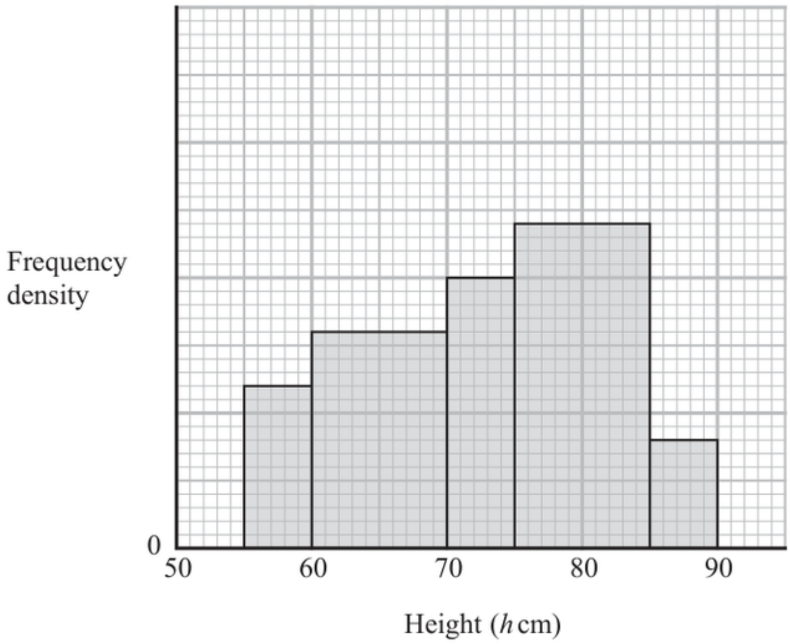
PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Histograms**

Jan 2022 P2H - Jan 2022 | P2H

20 The histogram gives information about the heights, h cm, of some tomato plants.



There are 12 tomato plants for which $75 < h \leqslant 85$
One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Histograms**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Use histogram

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

2. Use frequency table

For $75 < h \leq 85$, the frequency is 12 and the class width is 10, so

$$\text{frequency density} = \frac{12}{10} = 1.2$$

3. Use histogram

Using this scale from the histogram, the frequency densities are:

$$0.6,\ 0.8,\ 1.0,\ 1.2,\ 0.4$$

4. The total number of tomato

The total number of tomato plants is

$$(5)(0.6) + (10)(0.8) + (5)(1.0) + (10)(1.2) + (5)(0.4)$$

$$= 3 + 8 + 5 + 12 + 2 = 30$$

5. Solve inequality

For $h > 82.5$:

$$(2.5)(1.2) + (5)(0.4) = 3 + 2 = 5$$

6. Calculate probability

So the probability is

$$\frac{5}{30} = \frac{1}{6}$$

FINAL ANSWER

$\frac{1}{6}$

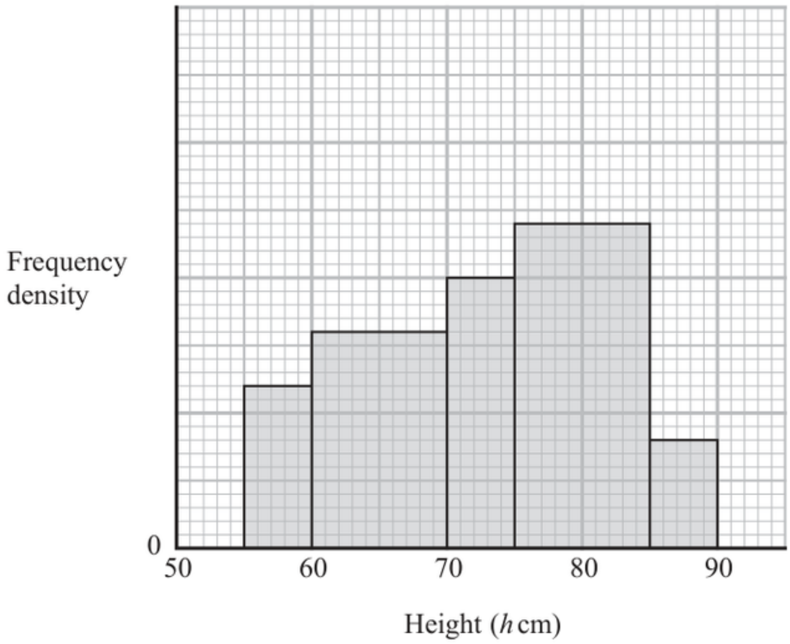
PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Histograms**

Jan 2022 P2H - Jan 2022 | P2H

20 The histogram gives information about the heights, h cm, of some tomato plants.



There are 12 tomato plants for which $75 < h \leqslant 85$
One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION Q#: 20 Marks: 4

TOPIC: **Histograms**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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1. Use histogram

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

2. Use frequency table

For $75 < h \le 85$, the frequency is 12 and the class width is 10, so

$$\text{frequency density} = \frac{12}{10} = 1.2$$

3. Use histogram

Using this scale from the histogram, the frequency densities are:

$$0.6,\ 0.8,\ 1.0,\ 1.2,\ 0.4$$

4. The total number of tomato

The total number of tomato plants is

$$(5)(0.6) + (10)(0.8) + (5)(1.0) + (10)(1.2) + (5)(0.4)$$

$$= 3 + 8 + 5 + 12 + 2 = 30$$

5. Solve inequality

For $h > 82.5$:

$$(2.5)(1.2) + (5)(0.4) = 3 + 2 = 5$$

6. Calculate probability

So the probability is

$$\frac{5}{30} = \frac{1}{6}$$

FINAL ANSWER

$\frac{1}{6}$

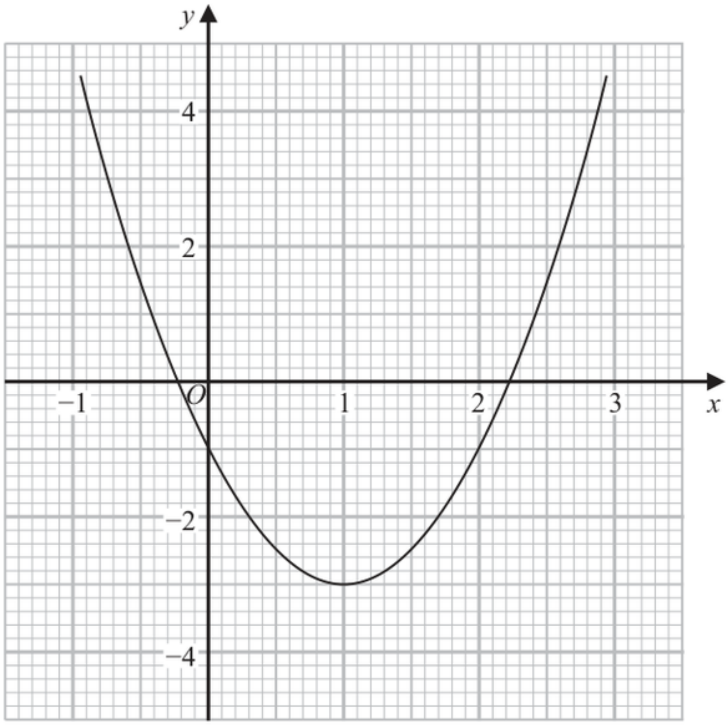
PAST PAPER QUESTION

Q#: 21 Marks: 5

Topic: **Graphs of Functions**

Jan 2022 P2H - Jan 2022 | P2H

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



(a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

....
(2)

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

....
(3)

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Graphs of Functions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Read the graph

For part (a), solve

$$2x^2 - 4x - 1 = 0$$

2. Read the graph

Using the graph gives approximately

$$x = -0.2, \quad x = 2.2$$

3. Read the graph

For part (b), we need to solve

$$x^2 - x - 1 = 0$$

4. Multiply by 2

Multiply by 2:

$$2x^2 - 2x - 2 = 0$$

5. The given graph is

The given graph is

$$y = 2x^2 - 4x - 1$$

6. Read the graph

Now

$$2x^2 - 2x - 2 = (2x^2 - 4x - 1) + 2x - 1$$

7. Read the graph

So draw the straight line

$$y = 1 - 2x$$

8. The intersections of this line

The intersections of this line with the curve give approximately

$$x = -0.6, \quad x = 1.6$$

FINAL ANSWER

(a) $-0.2,\ 2.2$; (b) draw $y = 1 - 2x$, giving $-0.6,\ 1.6$

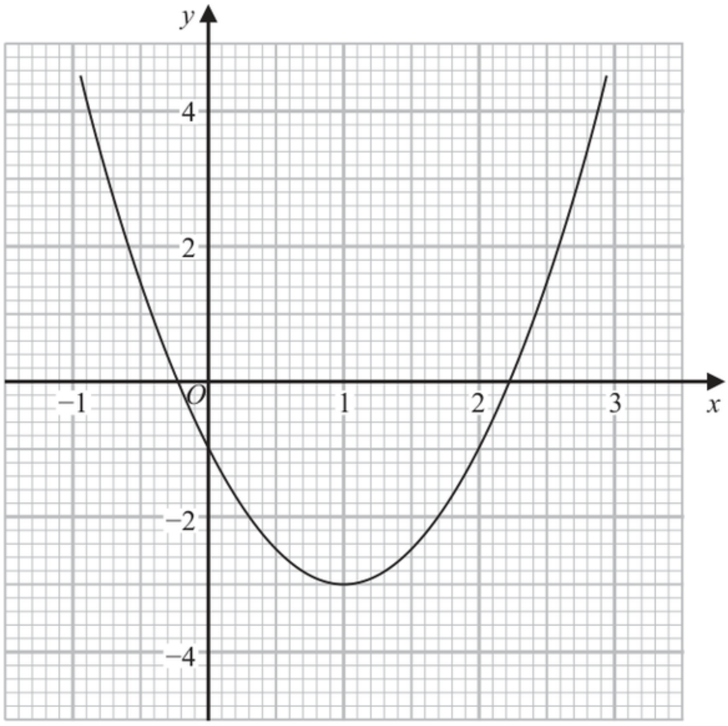
PAST PAPER QUESTION

Q#: 21 **Marks: 5**

TOPIC: **Graphs of Functions**

Jan 2022 P2H - Jan 2022 | P2H

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



(a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

....

(2)

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

....

(3)

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 **Marks: 5**

TOPIC: **Graphs of Functions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Read the graph

For part (a), solve

$$2x^2 - 4x - 1 = 0$$

2. Read the graph

Using the graph gives approximately

$$x = -0.2, \quad x = 2.2$$

3. Read the graph

For part (b), we need to solve

$$x^2 - x - 1 = 0$$

4. Multiply by 2

Multiply by 2:

$$2x^2 - 2x - 2 = 0$$

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$$2x^2 - 2x - 2 = (2x^2 - 4x - 1) + 2x - 1$$

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So draw the straight line

$$y = 1 - 2x$$

8. The intersections of this line

The intersections of this line with the curve give approximately

$$x = -0.6, \quad x = 1.6$$

FINAL ANSWER

(a) $-0.2,\ 2.2$; (b) draw $y = 1 - 2x$, giving $-0.6,\ 1.6$

PAST PAPER QUESTION Q#: 22 Marks: 5

TOPIC: **Solving Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

22 Here is a rectangle.

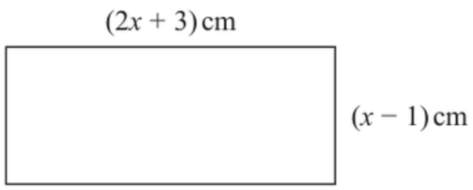

Diagram **NOT** accurately drawn

Given that the area of the rectangle is less than 75 cm^2

find the range of possible values of x

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Solving Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The rectangle has side lengths

The rectangle has side lengths

$$2x + 3$$

2. Use Functions Solving Inequalities

and

$$x - 1$$

3. Solve inequality

For the side lengths to be possible,

$$x > 1$$

4. The area is less than

The area is less than 75:

$$(2x + 3)(x - 1) < 75$$

$$2x^2 + x - 3 < 75$$

$$2x^2 + x - 78 < 0$$

5. Factorise

Factorise:

$$2x^2 + x - 78 = (2x + 13)(x - 6)$$

6. Evaluate fraction

So

$$-\frac{13}{2} < x < 6$$

7. Solve inequality

Combining with $x > 1$:

$$1 < x < 6$$

FINAL ANSWER

$1 < x < 6$

PAST PAPER QUESTION

Q#: 22 **Marks: 5**

TOPIC: **Solving Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

22 Here is a rectangle.

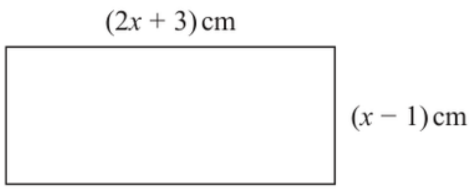

Diagram **NOT** accurately drawn

Given that the area of the rectangle is less than $75\,\text{cm}^2$

find the range of possible values of x

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Solving Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The rectangle has side lengths

The rectangle has side lengths

$$2x + 3$$

2. Use Functions Solving Inequalities

and

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For the side lengths to be possible,

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The area is less than 75:

$$(2x + 3)(x - 1) < 75$$

$$2x^2 + x - 3 < 75$$

$$2x^2 + x - 78 < 0$$

5. Factorise

Factorise:

$$2x^2 + x - 78 = (2x + 13)(x - 6)$$

6. Evaluate fraction

So

$$-\frac{13}{2} < x < 6$$

7. Solve inequality

Combining with $x > 1$:

$$1 < x < 6$$

FINAL ANSWER

$1 < x < 6$

PAST PAPER QUESTION

Q#: 23 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2H - Jan 2022 | P2H

23 The diagram shows triangle PQR

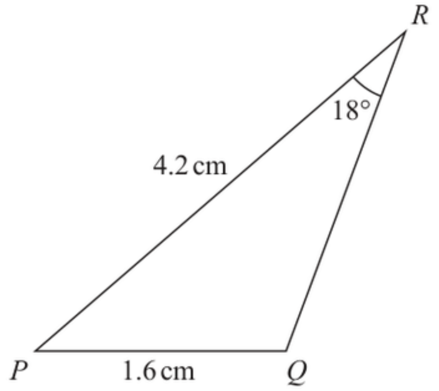


Diagram **NOT** accurately drawn

$PQ = 1.6$ cm $\quad PR = 4.2$ cm $\quad$ Angle $PRQ = 18°$

Given that angle PQR is obtuse,

work out the area of triangle PQR
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION *Dr. Eslam Ahmed*

1. Use trigonometry

In triangle PQR,

$$PQ = 1.6, \qquad PR = 4.2, \qquad \angle PRQ = 18^\circ$$

2. Use the sine rule

Use the sine rule:

$$\frac{\sin Q}{PR} = \frac{\sin R}{PQ}$$

$$\frac{\sin Q}{4.2} = \frac{\sin 18^\circ}{1.6}$$

$$\sin Q = \frac{4.2 \sin 18^\circ}{1.6}$$

$$\sin Q = 0.811169\ldots$$

3. Use trigonometry

Since angle PQR is obtuse,

$$Q = 180^\circ - \sin^{-1}(0.811169\ldots)$$

$$Q = 125.7896\ldots^\circ$$

4. Use trigonometry

Now find angle P:

$$P = 180^\circ - 18^\circ - 125.7896\ldots^\circ$$

$$P = 36.2103\ldots^\circ$$

5. Area of triangle

Area of triangle PQR:

$$\frac{1}{2}(PQ)(PR)\sin P$$

$$= \frac{1}{2}(1.6)(4.2)\sin(36.2103\ldots^\circ)$$

$$= 1.984925\ldots$$

6. Correct to 3 significant figures

Correct to 3 significant figures:

$$1.98$$

FINAL ANSWER

$\boxed{1.98\ \text{cm}^2}$

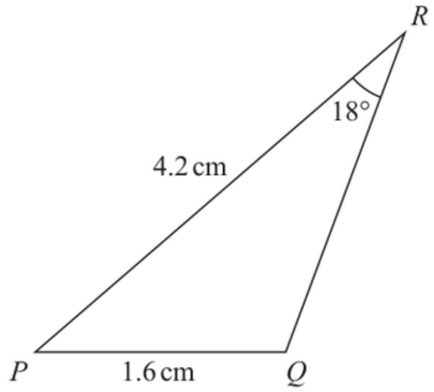
PAST PAPER QUESTION

Q#: 23 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2H - Jan 2022 | P2H

23 The diagram shows triangle PQR



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Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION *Dr. Eslam Ahmed*

1. Use trigonometry

In triangle PQR,

$$PQ = 1.6, \qquad PR = 4.2, \qquad \angle PRQ = 18^\circ$$

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Use the sine rule:

$$\frac{\sin Q}{PR} = \frac{\sin R}{PQ}$$

$$\frac{\sin Q}{4.2} = \frac{\sin 18^\circ}{1.6}$$

$$\sin Q = \frac{4.2 \sin 18^\circ}{1.6}$$

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Since angle PQR is obtuse,

$$Q = 180^\circ - \sin^{-1}(0.811169\ldots)$$

$$Q = 125.7896\ldots^\circ$$

4. Use trigonometry

Now find angle P:

$$P = 180^\circ - 18^\circ - 125.7896\ldots^\circ$$

$$P = 36.2103\ldots^\circ$$

5. Area of triangle

Area of triangle PQR:

$$\frac{1}{2}(PQ)(PR)\sin P$$

$$= \frac{1}{2}(1.6)(4.2)\sin(36.2103\ldots^\circ)$$

$$= 1.984925\ldots$$

6. Correct to 3 significant figures

Correct to 3 significant figures:

$$1.98$$

FINAL ANSWER

$\boxed{1.98 \text{ cm}^2}$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Differentiation**

Jan 2022 P2H - Jan 2022 | P2H

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geqslant 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is a m/s^2

Find the value of a

$a =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION Q#: 24 Marks: 5

TOPIC: **Differentiation**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Split the ratio

$$x = 4t^3 - 27t + 8$$

2. Velocity is

Velocity is

$$\frac{dx}{dt} = 12t^2 - 27$$

3. The direction of motion reverses

The direction of motion reverses when velocity is zero:

$$12t^2 - 27 = 0$$

$$12t^2 = 27$$

$$t^2 = \frac{9}{4}$$

4. Split the ratio

Since $t \geq 0$, $t = \frac{3}{2}$.

5. Acceleration is

Acceleration is

$$\frac{d^2x}{dt^2} = 24t$$

6. Split the ratio

At $t = \frac{3}{2}$,

$$a = 24\left(\frac{3}{2}\right) = 36$$

FINAL ANSWER

$a = 36$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Differentiation**

Jan 2022 P2H - Jan 2022 | P2H

24 A particle P moves along a straight line that passes through the fixed point O

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The acceleration of P at the instant when P is at A is a m/s^2

Find the value of a

$a =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Differentiation**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Split the ratio

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$$\frac{dx}{dt} = 12t^2 - 27$$

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The direction of motion reverses when velocity is zero:

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Since $t \geq 0$, $t = \frac{3}{2}$.

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Acceleration is

$$\frac{d^2x}{dt^2} = 24t$$

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At $t = \frac{3}{2}$,

$$a = 24\left(\frac{3}{2}\right) = 36$$

FINAL ANSWER

$a = 36$

PAST PAPER QUESTION

Q#: 25 Marks: 5

TOPIC: **Functions**

Jan 2022 P2H - Jan 2022 | P2H

25 The function g is defined as

$$g : x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x : x \geqslant 3\}$$

(a) Express the inverse function g^{-1} in the form $g^{-1} : x \mapsto \ldots$

$g^{-1} : x \mapsto$ ....

(4)

(b) State the domain of g^{-1}

....

(1)

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION Q#: 25 Marks: 5

TOPIC: **Functions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

$$g(x) = 5 + 6x - x^2$$

2. Complete the square

Complete the square:

$$g(x) = 14 - (x - 3)^2$$

3. Let

Let

$$y = 14 - (x - 3)^2$$

$$(x - 3)^2 = 14 - y$$

4. Find inverse function

The domain of g is $x \geq 3$, so take the positive square root:

$$x - 3 = \sqrt{14 - y}$$

$$x = 3 + \sqrt{14 - y}$$

5. Simplify surd

Therefore

$$g^{-1} : x \mapsto 3 + \sqrt{14 - x}$$

6. Calculate statistic

The range of g is $x \leq 14$, so this is the domain of g^{-1}.

FINAL ANSWER

(a) $g^{-1} : x \mapsto 3 + \sqrt{14 - x}$, (b) domain $x \leq 14$

PAST PAPER QUESTION Q#: 25 Marks: 5

TOPIC: **Functions**

Jan 2022 P2H - Jan 2022 | P2H

25 The function g is defined as

$$g : x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x : x \geqslant 3\}$$

(a) Express the inverse function g^{-1} in the form $g^{-1} : x \mapsto \ldots$

$g^{-1} : x \mapsto$ ....

(4)

(b) State the domain of g^{-1}

....

(1)

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 5

TOPIC: **Functions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

$$g(x) = 5 + 6x - x^2$$

2. Complete the square

Complete the square:

$$g(x) = 14 - (x - 3)^2$$

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The domain of g is $x \geq 3$, so take the positive square root:

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$$x = 3 + \sqrt{14 - y}$$

5. Simplify surd

Therefore

$$g^{-1} : x \mapsto 3 + \sqrt{14 - x}$$

6. Calculate statistic

The range of g is $x \leq 14$, so this is the domain of g^{-1}.

FINAL ANSWER

(a) $g^{-1} : x \mapsto 3 + \sqrt{14 - x}$, (b) domain $x \leq 14$

PAST PAPER QUESTION

Q#: 26 Marks: 5

TOPIC: **Sequences**

Jan 2022 P2H - Jan 2022 | P2H

26 An arithmetic series has first term a and common difference d, where d is a prime number.

The sum of the first n terms of the series is S_n and

$$S_m = 39$$
$$S_{2m} = 320$$

Find the value of d and the value of m
Show clear algebraic working.

$d =$ ....

$m =$ ....

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION Q#: 26 Marks: 5

TOPIC: **Sequences**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

$$S_m = 39$$

$$S_{2m} = 320$$

2. Use

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

3. Calculate value

Then

$$2S_m = m(2a + (m-1)d) = 78$$

4. Calculate value

and

$$S_{2m} = m(2a + (2m-1)d) = 320$$

5. Subtract

Subtract:

$$S_{2m} - 2S_m = m[(2m-1)d - (m-1)d]$$

$$320 - 78 = m(md)$$

$$242 = m^2 d$$

$$242 = 2 \times 11^2$$

6. Square factor

Since d is prime and m^2 is a square factor,

$$d = 2, \quad m^2 = 121$$

$$m = 11$$

FINAL ANSWER

$d = 2,\ m = 11$

PAST PAPER QUESTION

Q#: 26 Marks: 5

TOPIC: **Sequences**

Jan 2022 P2H - Jan 2022 | P2H

26 An arithmetic series has first term a and common difference d, where d is a prime number.

The sum of the first n terms of the series is S_n and

$$S_m = 39$$

$$S_{2m} = 320$$

Find the value of d and the value of m
Show clear algebraic working.

$d =$ ....

$m =$ ....

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 5

TOPIC: **Sequences**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

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$$S_{2m} = 320$$

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Subtract:

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$$320 - 78 = m(md)$$

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$$242 = 2 \times 11^2$$

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Since d is prime and m^2 is a square factor,

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FINAL ANSWER
$d = 2,\ m = 11$